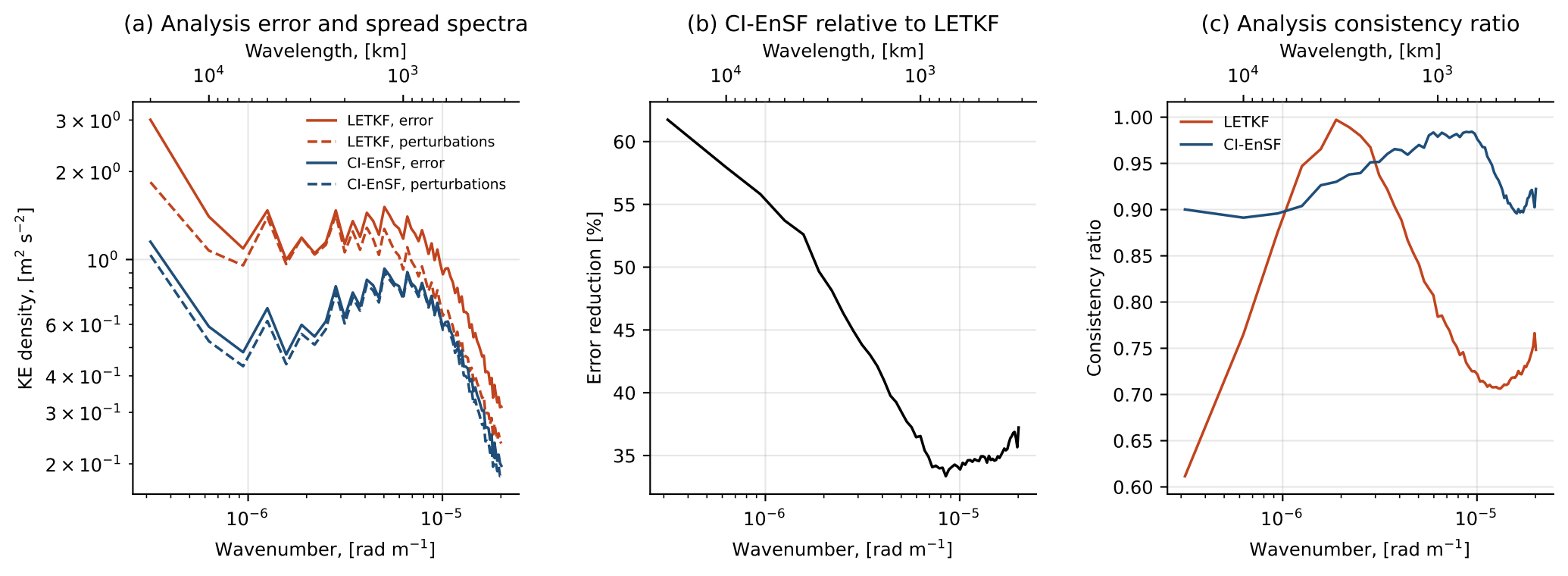
(a) Analysis error and spread spectra
Wavelength, [km]
$10^4$
$10^3$
$3 \times 10^0$
$2 \times 10^0$
$10^0$
$6 \times 10^{-1}$
$4 \times 10^{-1}$
$3 \times 10^{-1}$
$2 \times 10^{-1}$
KE density, [m$^2$ s$^{-2}$]
LETKF, error
LETKF, perturbations
CI-EnSF, error
CI-EnSF, perturbations
$10^{-6}$
$10^{-5}$
Wavenumber, [rad m$^{-1}$]
(b) CI-EnSF relative to LETKF
Wavelength, [km]
$10^4$
$10^3$
60
55
50
45
40
35
Error reduction [%]
$10^{-6}$
$10^{-5}$
Wavenumber, [rad m$^{-1}$]
(c) Analysis consistency ratio
Wavelength, [km]
$10^4$
$10^3$
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
Consistency ratio
LETKF
CI-EnSF
$10^{-6}$
$10^{-5}$
Wavenumber, [rad m$^{-1}$]